Comparing Surface and Intramuscular Electromyography for Simultaneous and Proportional Control Based on a Musculoskeletal Model: A Pilot Study

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***Abstract*— Simultaneous and proportional control (SPC) of neural-machine interfaces uses magnitudes of smoothed electromyograms (EMG) as control inputs. Though surface EMG (sEMG) electrodes are common for clinical neural-machine interfaces, intramuscular EMG (iEMG) electrodes may be indicated in some circumstances (e.g., for controlling many degrees of freedom). However, differences in signal characteristics between sEMG and iEMG may influence SPC performance. We conducted a pilot study to determine the effect of electrode type (sEMG and iEMG) on real-time task performance with SPC based on a novel 2-degree-of-freedom EMG-driven musculoskeletal model of the wrist and hand. Four able-bodied subjects and one transradial amputee performed a virtual posture matching task with either sEMG or iEMG. There was a trend of better task performance with sEMG than iEMG for both able-bodied and amputee subjects, though the difference was not statistically significant. Thus, while iEMG may permit targeted recording of EMG, its signal characteristics may not be as ideal for SPC as those of sEMG. The tradeoff between recording specificity and signal characteristics is an important consideration for development and clinical implementation of SPC for neural-machine interfaces.**

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***Index Terms*— Musculoskeletal, model, prosthesis, control, virtual reality, electrodes.**

I. Introduction

NEURAL-MACHINE interfaces (NMI) based on electromyograms (EMG) have been widely used in clinical applications, such as prosthesis control and neurorehabilitation, and have potential application in teleoperation, gaming, and virtual reality. In some NMI applications, such as prosthesis control, one key control goal is to closely align the prosthesis output movement with the user's intended movement [1]. Achieving this goal remains a major challenge since healthy human movement is extremely complex, involving simultaneous, coordinated, continuously variable, and independently modulated motion at multiple joints. Most commercially-available prostheses only reproduce a small portion (e.g. direction of movement for one degree of freedom (DOF)) of the movement intent [2].

Algorithms have evolved considerably to estimate more movement intent characteristics from EMG. Conventional dual-site control, still widely used today, enables direct switching between two directions of movement for one DOF at a time, but requires users to switch DOFs manually [2], [3]. Machine learning-based algorithms, such as pattern recognition, allows users to switch among classes (involving either single DOF [4], [5] or simultaneous multi-DOF [6] movements) directly and intuitively. However, many pattern recognition-based controllers employ a single classifier that can select only one class at a time, so that prosthesis movements must be executed sequentially. To overcome this limitation, researchers introduced algorithms enabling simultaneous and proportional control (SPC) of multiple DOFs. Such algorithms include parallel dual-site control [7], multi-classifier or parallel pattern recognition [8], [9], non-negative matrix factorization [10], linear regression [11], and EMG energy orthonormalization [12]. Our group [13], [14] and others [15] recently introduced EMG-driven controllers based on musculoskeletal models and forward dynamics simulation.

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Compared to other algorithms, SPC algorithms replicate more human movement characteristics, including the ability to independently modulate motion at multiple joints simultaneously.

The number and type of DOFs that can be included in SPC is limited by the reliable accessibility of functionally distinct EMG signals. A straightforward way to record functionally distinct EMG is to measure it from individual muscles, since each muscle's biomechanical contribution to movement is unique. EMG for prosthesis control is recorded almost exclusively using surface (sEMG) electrodes placed on the skin over target muscles. Many of the reported SPC algorithms, including our model-based controller, have generally demonstrated good movement estimation with sEMG for up to three DOFs [11], [16], [17]. Though sEMG electrodes are convenient to place and non-invasive, their location and relatively large inter-pole spacing (i.e. large recording volume) may make it challenging or impossible to record muscle-specific EMG under some conditions. For instance, muscles for some DOF pairs are anatomically adjacent or overlapping, degrading individual muscles' EMG signal fidelity. In people with amputation, especially those with a short residual limb, parts of muscles that were once superficial may be smaller and deep to other muscles (i.e. less accessible) in the residual limb. Finally, muscles may shift relative to the overlying skin with limb posture changes, which may reduce the reliability of access to those muscles and, thus, movement intent estimation accuracy [18], [19]. These issues become more problematic as the number of desired DOFs for real-time control increases.

Compared to sEMG, intramuscular (iEMG) electrodes potentially offer greater, more precise access to muscle-specific EMG. iEMG electrodes are placed directly within the muscle belly and have relatively small inter-pole spacing, enabling more precise EMG recording. Additionally, since iEMG electrodes are placed within the muscle, their location may be more impervious to limb posture changes than sEMG electrodes. Given their advantages over sEMG, iEMG may improve SPC movement estimation accuracy or permit a higher number of DOFs for SPC. iEMG has been used to decode simultaneous, dexterous movements of the hand, including individual finger, wrist, and thumb movements in able-bodied subjects [20]–[22]. Implantable intramuscular electrodes are being developed (e.g. Ripple Neuro, USA; [23]) and would make chronic iEMG recording practical for clinical prosthesis control in the future.

Despite the potential advantages of iEMG electrodes for recording muscle-specific EMG, differences between sEMG and iEMG may affect the output of SPC. For instance, compared to sEMG, iEMG signals contain fewer, more distinct action potentials recorded from fewer motor units. The different way the electrodes interface with the user may also affect electrode motion artifact and user comfort. Others have compared sEMG and iEMG for real-time control based on pattern recognition, which extracts multiple time- and frequency-domain features from EMG to estimate movement intent [17], [24]. However, SPC algorithms process EMG in an entirely different way, usually computing the magnitude of the EMG (e.g. by computing the mean absolute value [11]) as an estimate of muscle contraction magnitude. For our real-time controller based on an EMG-driven musculoskeletal model, we rectify and smooth the EMG to estimate its magnitude [14]. Though SPC algorithms have previously been tested with iEMG, to our knowledge, a direct comparison of control performance between sEMG and iEMG has not been reported for any SPC algorithm. We previously tested our model for real-time control with sEMG [14], but not with iEMG.

The goal of our pilot study was to determine the effect of EMG recording method (sEMG vs iEMG) on real-time SPC based on our novel EMG-driven 2-DOF musculoskeletal model. In the current study, four able-bodied subjects and one transradial amputee performed a real-time control task in which they aligned a virtual hand with target postures on a computer screen. We hypothesized that task performance would be better for iEMG than sEMG across all test conditions. The proposed study indicated trends in differences in SPC performance between electrode type, which could influence the development and implementation of SPC for NMI.

II. Methods

We recruited 4 able-bodied subjects (AB1-AB4, ages 21-25 years) and one subject (TR) with transradial amputation. Subject TR (female, age 23) sustained a right transradial amputation during an ATV accident 19 months before participating in the study. Subject TR regularly used a myoelectric prosthesis with a dual-site myoelectric control scheme in daily life, and her residual forearm was approximately 1/3 of that of her intact forearm. All subjects otherwise had no history of chronic illness, neurological disorder, or recent musculoskeletal injury. Two subjects (AB1 and AB2) had experience using the model-based controller in a previous study [14]; all other subjects were naïve to the controller. The experimental protocol was approved by the University of North Carolina at Chapel Hill Institutional Review Board. All subjects gave their informed consent to participate.

A. Model Description

We used a previously developed controller that enabled SPC of wrist and metacarpophalangeal (MCP) flexion and extension based on a simplified planar two-DOF musculoskeletal model [13]. The model included 4 virtual muscles crossing either the wrist only or the wrist and MCP joints in agonist-antagonist pairs. The model's anthropometric and inertial properties approximated those of the human hand.

Using a constrained global numerical optimization procedure, as described in [13], we computed the values of twenty-two model parameters (6 for each of 2 bi-articular muscles crossing the wrist and MCP joints, 5 for each of 2 uni-articular muscles crossing the wrist only) that influence the virtual muscles' biomechanical contributions to wrist and MCP movements: optimal muscle length, *in situ* muscle length in neutral posture (0° wrist and MCP flexion/extension angle), maximum isometric force, moment arm about the wrist and MCP joints, and parallel elastic element stiffness. For AB1 and AB2, the parameters were computed using each subject's own experimental kinematic and sEMG data during isolated and

combined wrist and MCP flexion/extension movements in one static upper limb posture (arm and forearm neutral, elbow flexed to 90°). Models used by the remaining subjects (AB3, AB4, TR) were prescribed generic parameter values that were computed by averaging the optimized parameters from data collected across 6 able-bodied subjects and 9 static upper limb testing postures [25]. The model parameters for each subject have been included with the manuscript as supplementary data.

The model was implemented for real-time EMG-driven control of a planar stick figure virtual hand in MATLAB (Mathworks, Inc., USA) [14]. Muscle activations ranging from 0% (inactive muscle) to 100% (maximally active muscle) were computed from recorded EMG, as described below, and applied to the virtual muscles during a real-time forward dynamics simulation to estimate wrist and MCP joint angles. The posture of the virtual hand, which included the same DOFs as the musculoskeletal model, was displayed on a computer screen and updated at 20 Hz.

B. EMG Recording

In two separate test sessions, EMG used to control the virtual hand were recorded using either sEMG or iEMG electrodes. Subjects AB1, AB2, and AB3 tested with sEMG first, while AB4 and TR tested with iEMG first.

Before placing sEMG electrodes, the skin was cleaned with an alcohol pad. Four bipolar adhesive surface EMG electrodes (Norotrode 20, Myotronics Inc., USA), which had an electrolyte gel to reduce skin-electrode impedance, were placed approximately over the extensor carpi radialis longus, extensor digitorum, flexor carpi radialis, and flexor digitorum superficialis muscles. sEMG electrode placement was carefully performed by an experienced researcher by muscle palpation. For the amputee subject, sEMG was first attempted using dry electrodes (SX230, Biometrics Ltd, UK) so that they could be easily relocated to achieve the desired placement. The electrodes were connected to a preamplifier (MA-420, Motion Lab Systems, Inc., USA), and EMG signals were recorded at 1000 Hz (Motion Lab Systems, Inc., USA).

The skin near the iEMG electrode insertion site was first cleaned thoroughly with an alcohol pad. An ultrasound system (ECO 3, Chison, China) and linear transducer (L7S-A, Chison, China) were first used to identify the muscle of interest while the subject performed experimenter-directed isolated movements of either the wrist or MCP joints. Under ultrasound guidance, a pair of 0.002 in. Stablohm 650 wires with PTFE insulation (California Fine Wire Co, USA) were inserted percutaneously into the muscle belly using a 27G hypodermic needle. Insulation was removed by flame from approximately 1 mm of the inserted end and 1 cm of the recording end of both wires before the procedure. Electrodes were inserted into the muscle bellies of extensor carpi radialis longus, extensor digitorum, flexor carpi radialis, and flexor digitorum superficialis. The uninserted wire ends were secured to posts on a bipolar preamplifier (MA-416, Motion Lab Systems, Inc., USA). EMG was recorded at 1000 Hz (Motion Lab Systems, Inc., USA).

To confirm that electrodes were placed over or within the appropriate muscles, EMG were visually inspected while subjects performed experimenter-directed movements in each direction of movement (e.g. wrist flexion) corresponding to each targeted muscle. Electrodes were replaced if (1) EMG was inactive during a movement to which the target muscle is expected to contribute, or (2) there was high activity at a muscle (e.g. wrist extensor) during a movement that was not expected to involve that muscle.

Both sEMG and iEMG signals were processed as follows:

1) 100 timepoints (100 ms) of EMG were sampled for each channel.
2) To minimize filter artifacts generated at the ends of the data windows, each 100 ms window was appended to the previous 19 sample windows (a total of 2000ms of accumulated EMG data).
3) The accumulated EMG were high-pass filtered with a cutoff frequency of 40 Hz using a 1st-order Butterworth zero-lag filter, rectified, and low-pass filtered with a cutoff frequency of 1 Hz using a 1st-order Butterworth zero-lag filter.
4) The EMG from the 2nd most recent 100 ms window were extracted from the accumulated, filtered EMG.
5) The extracted EMG were further smoothed and down-sampled from 1000 Hz to 100 Hz by computing the moving average over 70 timepoints in 10-point intervals.
6) EMG values were normalized by the maximum EMG recorded during maximum voluntary contractions (EMG-MVC) to estimate the muscle activations of the four virtual muscles.

C. Model Calibration and Real-Time Virtual Task

After placing EMG electrodes, subjects were seated in front of a computer screen and held their upper limb in a static posture (arm neutral, elbow flexed to 90°). EMG-MVC were recorded while subjects sequentially performed maximum voluntary contractions of wrist and MCP flexion and extension. Subjects then attempted to move the virtual hand to each of the four tested target hand postures (Fig. 1). An experimenter adjusted linear gains manually to scale the input (EMG) and output (joint angles) so that each subject could (1) reach all targets comfortably and (2) reach a neutral 'base' hand posture when relaxed.

Subjects were instructed to move the virtual hand from the neutral 'base' hand posture to one of four target hand postures (Fig. 1) as quickly as possible. To ensure that subjects achieved each posture in a controlled manner, subjects were required to hold the virtual hand within ±8° of the base posture for 0.25 consecutive seconds and within ±5° of the target posture for 0.5 consecutive seconds. The base and target postures turned green when the subject was within the acceptable range. Subjects were given up to 60 seconds to achieve each target.

In each trial, subjects attempted all target hand postures in a randomized order. Able-bodied subjects performed one familiarization trial and 4 test trials first with the forearm in neutral posture, then with the forearm pronated approximately 90°; Subject TR performed the task in the neutral forearm posture

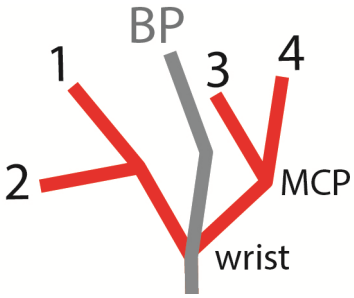

Fig. 1. Diagram of four target hand postures (red lines) and base hand posture (BP, gray line). The segments of the virtual hand (not shown) had the same dimensions as that of the target and base postures.

only since she was physically unable to maintain a pronated forearm posture. We recorded the EMG, muscle activations, and virtual hand kinematics for each trial.

D. Task Performance Measures

For each target, we computed several performance measures to quantify subjects' performance of the task. We computed the **task duration (TD)** or time from when the base posture was achieved to when the target posture was achieved. To evaluate how directly subjects moved from the base to target posture, we computed the **normalized path length (NPL)** as the ratio between the subjects' angular trajectory length and minimum angular trajectory length between base and target postures (1):

$$NPL = \frac{\sum_{k=1}^{n-1} \sqrt{\left(\theta_{k+1}^{w} - \theta_{k}^{w}\right)^2 + \left(\theta_{k+1}^{m} - \theta_{k}^{m}\right)^2}}{\sqrt{\left(\theta_{t}^{w} - \theta_{b}^{w}\right)^2 + \left(\theta_{t}^{m} - \theta_{b}^{m}\right)^2}} \quad (1)$$

In (1), n is the number of timepoints, θ^w and θ^m are the wrist and MCP joint angles, respectively, and θ_b and θ_t are the joint angles at the base and target postures, respectively. Finally, we computed the **number of overshoots (NO)** as the number of times that the virtual hand moved in and out of the target posture range.

Since TD and NPL are cumulative measures of performance, we divided TD and NPL among different spatial regions to see how these measures were distributed over the task. Regions were defined by the time when the subjects first passed different distance-to-target thresholds. Distance was measured in joint angle space.

E. Data Analysis

To compare able-bodied task performance among test conditions – EMG (sEMG, iEMG), Posture (neutral forearm, pronated forearm), and Target (1-4) - we used the following statistical model (2).

$$\begin{aligned} B_h \sim N y_{hijkl} = \mu + B_h + \alpha_i + E_{hi}^{W} + \beta_j + (\alpha\beta)_{ij} \\ + E_{hij}^{S} + \gamma_k + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} \\ + E_{hijk}^{SS} + E_{hijkl}.(0, \sigma_b^2) \end{aligned} \quad (2)$$

In (2), y_{hijkl} denotes the measurement from the l-th trial ($l = 1, \ldots, 4$) for the experimental setup for subject h with EMG i, Posture j, and Target k. Here μ represents some overall effect, B_h is a random effect for the hth subject, α_i is the overall effect of the ith EMG, β_j is the overall effect of the jth Posture, and γ_k is the overall effect of the kth Target. Two-factor interactions between the latter three effects are indicated by parentheses (e.g. $(\alpha\beta)_{ij}$). The random effects E_{hi}^{W}, E_{hij}^{S}, and E_{hijk}^{SS} are the experimental error terms representing variation between individual trials for a given experimental setup.

We varied the above model slightly based on the form of the performance metrics. Since NO is a count variable, we used a Poisson regression model with a form identical to (2) (no residual error term). For TD and NPL, a log transformation was applied to the measured values in order to satisfy the normality requirement of the residual error term. All analyses were performed using PROC GLIMMIX in SAS 9.4. Post-hoc pairwise comparisons were made to investigate significant main and interaction effects. We did not make statistical comparisons between sEMG and iEMG for subject TR since the data were recorded from only one subject. Due to the low number of subjects and power of our analysis, we used $\alpha = 0.10$ as our cutoff for statistical significance.

The model permits subjects to control wrist and MCP motion simultaneously. We computed a ratio of wrist-to-total angular velocity (r_ω) at each timepoint to determine the extent to which subjects moved the virtual wrist and MCP joints simultaneously during the task (3).

$$r_\omega = \frac{\omega_{wrist}}{\omega_{wrist} + \omega_{MCP}} \quad (3)$$

To compare simultaneous control between sEMG and iEMG trials, we computed the distribution of r_ω across timepoints for each trial, then computed the mean distribution across trials for each electrode type.

We compared raw and processed EMG signal characteristics to see whether there were differences between signals that could have affected subjects' task performance. Between pairs of trials of the same subject, target, and forearm posture, we qualitatively compared the raw EMG data, normalized power spectral density function, mean and median power frequencies [26], and muscle activations (i.e. processed, normalized EMG). Normalized power spectral density was computed for each trial as the average power spectral density across consecutive 1000ms windows and channels, normalized by the sum of power across all frequencies.

We plotted mean ± 1 standard deviation movement trajectory curves for each target for AB subjects to provide qualitative context to the quantitative performance measure data. We qualitatively compared subject TR's task performance measures and joint angle trajectory between sEMG and iEMG trials, since the data was collected from one TR subject.

III. Results

Across AB subjects, there was no statistically significant difference between EMG electrode types for TD ($p = 0.3311$), NPL ($p = 0.1129$), and NO ($p = 0.3594$) (Fig. 2). However, there was a trend of better performance with sEMG than with iEMG. Qualitatively, as in AB subjects, subject TR's

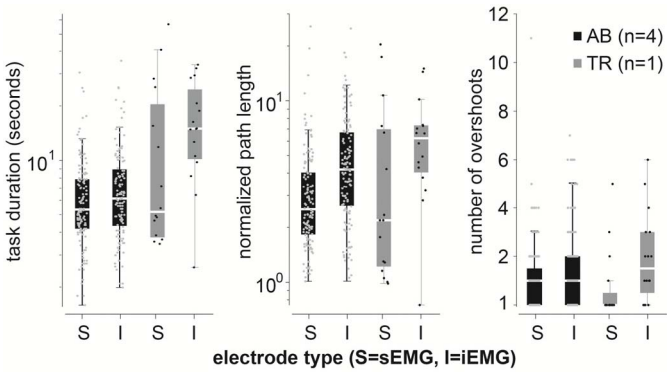

Fig. 2. Box plot comparison of task performance between sEMG and iEMG trials based on measurements of task duration (TD), normalized path length (NPL), and number of overshoots (NO). Data are for all trials for AB (two forearm postures) and TR (one forearm posture) subjects. Overlaid points show performance values for each trial. TD and NPL y-axes are plotted on a logarithmic scale.

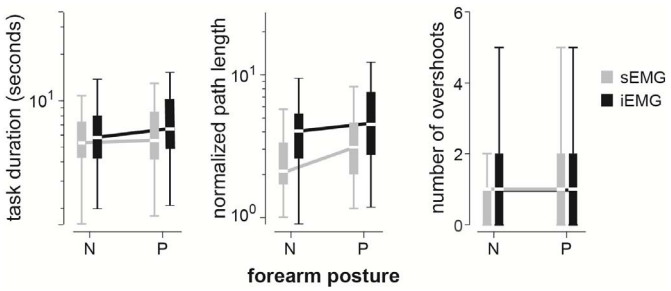

Fig. 3. Box plot comparison of task performance between neutral forearm posture (N) and pronated forearm posture (P) for both sEMG and iEMG. TD and NPL y-axes are plotted on a logarithmic scale. There was no statistically significant difference in task performance between forearm postures.

performance for sEMG was better than for iEMG, though performance was more variable for sEMG than for iEMG.

There was also no statistically significant difference in AB subjects' task performance between forearm postures for TD ($p = 0.2474$), NPL ($p = 0.1811$), and NO ($p = 0.2035$) (Fig. 3). The trend of differences between forearm postures was mixed; NPL was less impacted by forearm posture for iEMG than sEMG, but there was no visible difference in forearm posture effect on TD and NO between iEMG than sEMG.

For log-transformed variable TD, no effects were statistically significant. For log-transformed variable NPL, only the Target main effect ($p = 0.0011$) was significant at the 0.10 level. Pairwise comparisons of Target across the levels of EMG and Posture showed that target 3 ($\bar{x} = 2.36$, standard error (SE) $= 0.32$) had significantly lower NPL than targets 1 ($\bar{x} = 4.53$, SE $= 0.62$, $p < 0.001$), 2 ($\bar{x} = 3.68$, SE $= 0.50$, $p = 0.0053$) and 4 ($\bar{x} = 3.54$, SE $= 0.48$, $p = 0.0101$).

For variable NO, both target ($p = 0.0710$) and the EMG∗Target interaction ($p = 0.0641$) were significant at the 0.10 level. Based on post-hoc pairwise comparisons, there was no statistically significant difference between sEMG and iEMG for any of the targets. However, for sEMG, we found

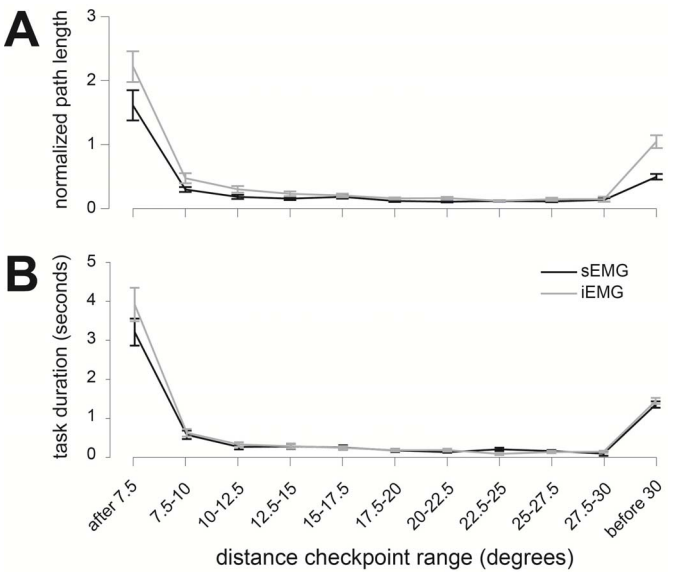

Fig. 4. Mean (±standard error) of (A) normalized path length (NPL) and (B) task duration (TD) within each distance checkpoint range for able-bodied subjects. For example, the range "7.5-10" indicates the period of time between when the virtual hand was first within 10 degrees and 7.5 degrees of the target posture, in joint angle space.

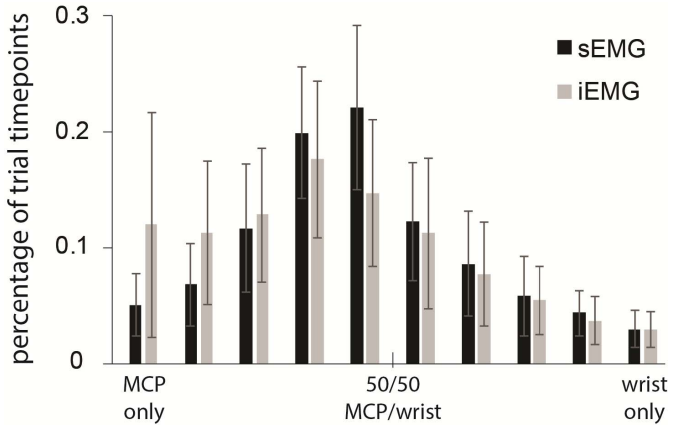

Fig. 5. Bar chart of mean (±1 standard deviation) joint angular velocity distribution across trials with sEMG and iEMG.

that target 2 had a higher NO ($\bar{x} = 1.83$, SE $= 0.45$) than the other targets ($0.73 \leq \bar{x} \leq 0.86$, $p = 0.0065$).

To evaluate the relative contributions of portion of the task to TD and NPL, we divided TD and NPL based on the timepoints when the virtual hand reached different distances from the target (Fig. 4). Across trials for able-bodied subjects, NPL and TD were highest at the end of the task when the virtual hand was near the target, suggesting that subjects could more easily move *toward* the target than stay *within* the target to complete the task. This same task portion - when the virtual hand was closest to the target - was also the task portion with the greatest difference in NPL and TD between sEMG and iEMG trials, indicating that it was more difficult for subjects to stay within the target with iEMG than with sEMG.

The average percentage of timepoints in which subjects preferentially moved the wrist was similar between sEMG and iEMG (Fig. 5). However, subjects tended to preferentially

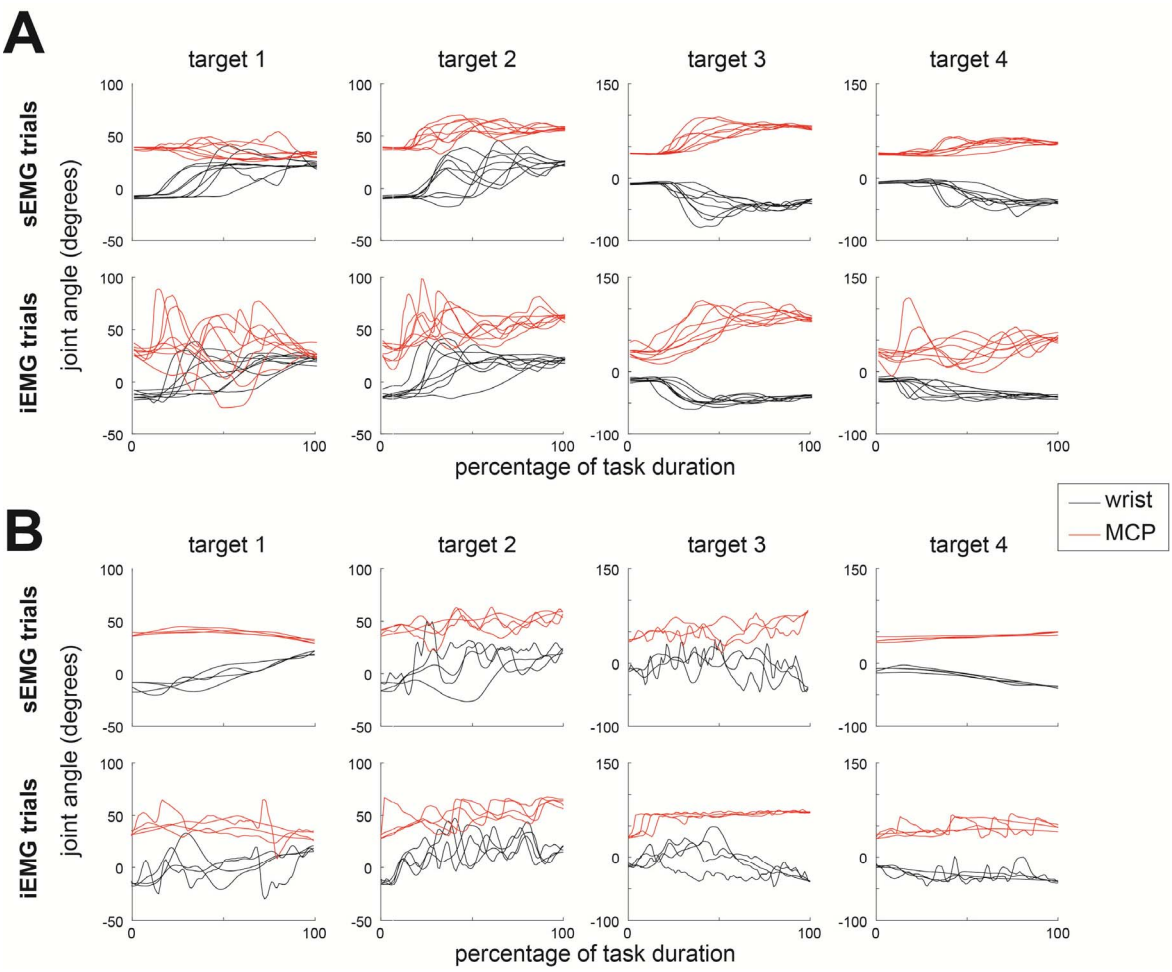

Fig. 6. Comparison of wrist and MCP angles for each target between sEMG and iEMG trials for (A) subject AB3 (both forearm postures, 8 trials per target), and (B) subject TR (one forearm posture, 4 trials per target).

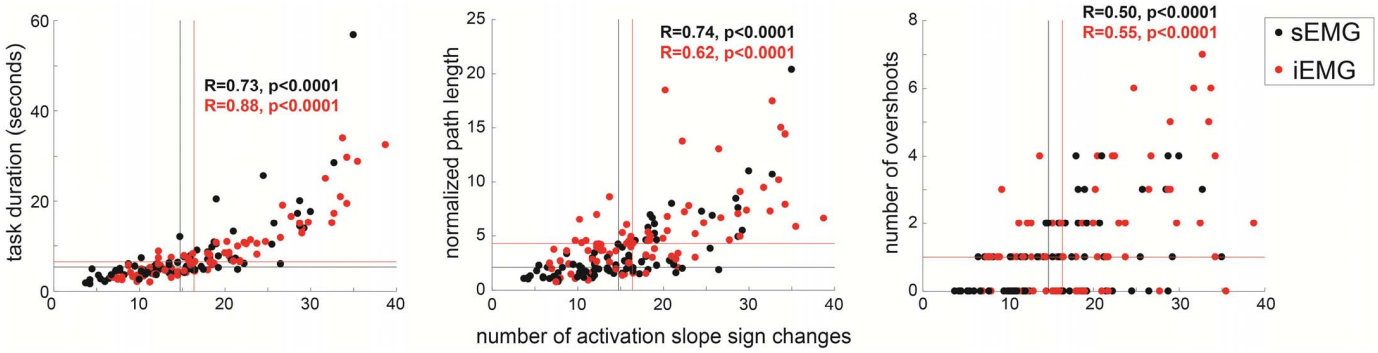

Fig. 7. Scatter plots of task duration (TD), normalized path length (NPL), and number of overshoots (NO) for each electrode type with respect to the number of activation slope sign changes in each trial. Horizontal and vertical lines indicate the median value for each electrode type (For the task duration figure at right, the median task duration for sEMG was zero). Data plotted for all subjects and trials.

move the MCP joint more, rather than coordinate MCP motion with wrist motion, with sEMG than with iEMG. In other words, there was a trend toward more simultaneous coordination of wrist and MCP movement with sEMG than with iEMG.

Qualitatively, there appeared to be more inter-trial variation and oscillation in the joint angle trajectories for iEMG than sEMG (Fig. 6). Such kinematic differences between electrode types could potentially explain the trend in task performance differences between electrode types. Since the control input to the model-based controller were muscle activations computed from EMG, we compared the task performance (control outcome) to the mean number of slope sign changes of the muscle activations (control input) per EMG channel for each trial. There was a statistically significant correlation ($p < 0.0001$) between number of activation slope sign changes and each

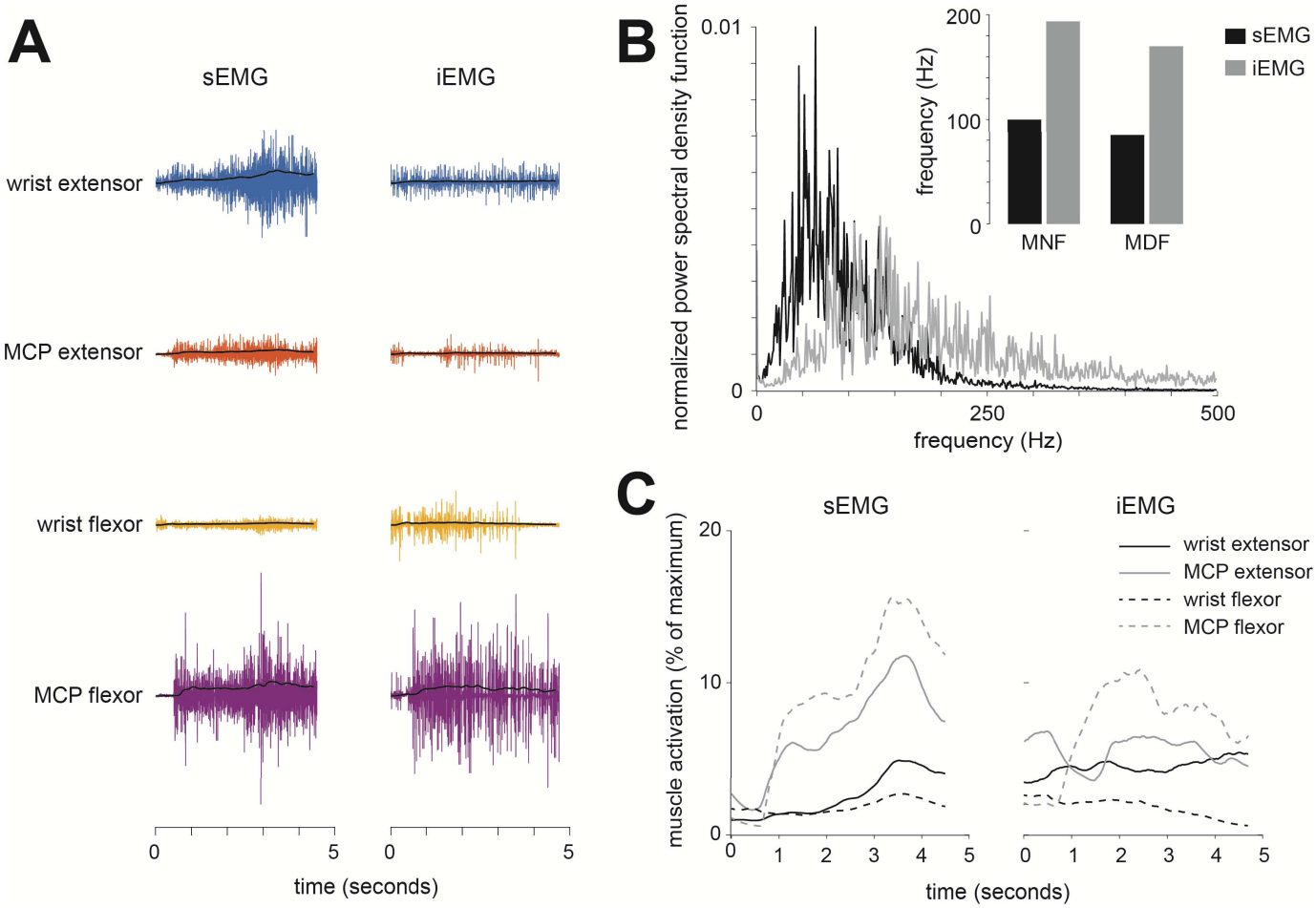

Fig. 8. Sample trial data comparison between trials with sEMG and iEMG (data from AB2, Target 3). (A) Raw and processed (black line) EMG data for each channel, labeled by the target muscle's primary biomechanical action. (B) Normalized power spectral density function of sEMG and iEMG during the trials. The bar chart shows mean power frequency (MNF) and median power frequency (MDF). (C) Muscle activations over the trial, computed as the processed EMG normalized by the maximum processed EMG value recorded during maximum voluntary contractions.

task performance measure (Fig. 7). Median values of all task performance measures and the number of activation slope sign changes were higher for iEMG than sEMG.

To investigate the potential source of differences in number of activation slope sign changes between electrode types, we compared raw and processed EMG signal characteristics between two exemplary trials, one each for iEMG and sEMG, with similar performance for the same subject and target. There appeared to be more high-frequency "spikes" in the iEMG than in the sEMG (Fig. 8a). This was confirmed by power density spectral analysis, which revealed lower mean and median power frequencies for sEMG than iEMG, as previously reported [27] (Fig. 8b). Despite the difference in raw EMG power distribution, the relative magnitudes of muscle activation were comparable (Fig. 8c).

IV. Discussion

We hypothesized that, for our musculoskeletal model-based EMG decoder, subjects' task performance would be better and more consistent between postures with iEMG electrodes than with sEMG electrodes. This is because iEMG can measure the required muscle-specific EMG more precisely and with less cross-talk than sEMG. However, there was a trend of better task performance with sEMG than iEMG, though differences were not statistically significant. Our detailed analysis of EMG and task performance indicated that there was a positive correlation between the number of activation slope sign changes during a trial and poorer task performance. Since estimated muscle activations were the inputs to the model-based controller, more oscillatory activations caused the virtual hand to oscillate more with iEMG than sEMG, likely contributing to the observed difference in task performance.

Though the observed trend was unexpected, there are several potential negative characteristics of iEMG that may have outweighed its advantage over sEMG in permitting more targeted EMG recording. Some characteristics are related to the signal. For instance, sEMG may contain more accurate information about whole-muscle force generation than iEMG since it records from a larger number of motor units. Characteristic iEMG signals contain more intermittent "spikes" and high-frequency content (e.g. Fig. 8) from which estimating the timing and magnitude of whole-muscle force generation may have been inaccurate. iEMG electrodes are also known to be highly sensitive to movement, and muscle movements

can introduce noise due to relative movement between the skin and muscle through which the electrodes are placed. Noisy or intermittent EMG signals, and subjects' actions to correct or overcome errors due to these signal characteristics, could have degraded performance and caused higher activation slope sign changes with iEMG. Regarding the electrodes themselves, able-bodied subjects expressed discomfort when moving the wrist and hand with the iEMG electrodes. With discomfort, subjects possibly performed the task more slowly (increasing TD) and less accurately (increasing NPL and NO), had more difficulty holding the virtual hand at the target (e.g Fig. 4), or were reluctant to coordinate wrist and MCP motions (e.g. Fig. 5).

Another possible explanation for the observed trend in task performance between electrode types is that the model may be inherently impervious to both cross-talk and EMG variation due to movement condition (e.g. posture) changes. The model's robustness may be, in part, because the estimated movements are based on the net biomechanical intent of 4 EMG signals (and their corresponding muscles in the model), rather than any one EMG signal. For instance, attempting wrist flexion may generate cross-talk leading to some unintended MCP flexion, but this could be overcome if the user simultaneously attempts MCP extension. Unlike algorithms that enable binary (i.e. either "on" or "off") movement estimates, the continuous movement estimates made by the model-based controller allow users to make slight adjustments in contraction magnitude to correct for position errors. We expect that the model's robustness with sEMG would diminish if more muscles and DOFs were added, though movement estimations with sEMG were good for one AB subject when our model was expanded to 3 DOFs (added pronation/supination) [16]. The model complexity for which iEMG becomes more advantageous should be investigated further in future studies.

A possible explanation for why task performance did not differ between forearm postures for either sEMG or iEMG is that the two tested forearm postures did not affect the electrode location and potential cross-talk vulnerability of sEMG as much as other conditions might (e.g. different upper extremity postures or dynamic upper extremity movements). Additionally, we used sEMG electrodes with adhesive tape and electrolyte gel, which improves adherence to the skin and decreases skin electrical impedance, respectively. Conversely, current commercially-available myoelectric prostheses use dry electrodes (e.g. MC TRIAD preamp, Motion Control, Inc., USA) that neither adhere to the skin nor use an electrolyte gel. Dry electrodes are convenient for chronic EMG recording and permit users to quickly don/doff the prosthetic socket [28], though they may more easily shift relative to the muscles with skin and socket movement during use [29]. Thus, the electrode shift with our adhesive sEMG electrodes may have been less than what dry, socket-embedded sEMG electrodes experience. To determine the effect of sEMG electrode type, we plan to test our model-based controller with conventional sockets and dry sEMG electrodes in future studies.

The subject with transradial amputation could complete the posture matching task and achieve all four target postures, but with task performance that was generally slower and less precise than that of able-bodied subjects. This is not surprising, since previous studies investigating EMG decoding algorithms for wrist and hand movement similarly reported poorer performance for subjects with amputation than for able-bodied subjects [30], [31]. There are several factors that may have contributed to the difference in task performance between able-bodied and amputee subjects. For instance, subject TR attempted MCP movements that she could not generate physiologically and were not required for using her personal myoelectric prostheses, and verbally reported difficulty sensing MCP movements. Disuse and reduced sensory feedback are thought to contribute to post-amputation neuromotor reorganization [32], [33], which may alter the motor commands generated during attempted wrist and hand movements. Training reportedly improves movement estimation accuracy and prosthesis control performance for other control algorithms [34]–[38]. Investigating the extent to which similar training methods could improve model-based control performance will be needed to promote the clinical translation of our model-based controller.

The observation that task performance was not better with iEMG is potentially advantageous for translating our 2-DOF musculoskeletal model-based controller for clinical application. sEMG is the more practical and widely used EMG recording method for myoelectric prostheses. Permanently implanted iEMG electrodes are being developed [23], [39], [40] but are not yet approved for clinical use. Transdermal fine-wire iEMG electrodes are widely used in clinical settings but are impractical for chronic EMG recording.

We tested subjects' performance of a posture matching task, a test paradigm used in other studies [41]. In some respects, the posture matching task was simpler than the path tracing task we used in our previous study [14], since subjects could focus solely on matching the target, rather than the target and trajectory. The path tracing task required relatively complex, dynamic coordination of the virtual hand's wrist and MCP joint movements to follow a path, whereas the path taken to match the target posture was not constrained. However, unlike in the path tracing task, subjects had to hold the virtual hand in the target for 0.5 consecutive seconds, which better mimics the motor control requirements of real-world grasping tasks for which hand posture stability is critical.

Individuals with amputation at the transradial level were a convenient population for testing our EMG decoder because they retain residual muscles that once contributed to wrist and hand movements. However, many individuals with higher level amputation do not retain these muscles, precluding direct recording of EMG associated with wrist and hand movements. Targeted muscle reinnervation is a promising surgical technique that, as previously demonstrated, can provide access to EMG associated with the wrist and hand in high-level amputees [42], [43]. Others have developed implantable nerve-cuff electrodes that interface directly with motor and sensory neurons [44], [45]. Using these advanced techniques, it may be possible to record muscle- or joint-specific neural signals for decoding simultaneous multi-joint movements of the wrist and hand for a wider segment of the amputee population.

There were several limitations of our study. We tested subjects' performance of a virtual task as they held a static upper limb posture, which differs from real-world tasks that may involve the amputee wearing a physical prosthesis, interacting with physical objects with mass, and performing dynamic upper limb movements such as reaching. We only tested the two EMG recording methods with one EMG decoding algorithm based on a musculoskeletal model, though several other algorithms have been developed for decoding wrist and hand movements from EMG [4], [5], [8], [11], [20]–[22], [30]; in future studies, we will test and compare the performance of our model-based EMG decoder with existing EMG decoding methods.

As this was a pilot study, we tested only 5 subjects, one of whom had a transradial amputation. The number of amputees that can be recruited at any one test site may be small. Many previous studies preliminarily evaluating SPC for prostheses included no amputee subjects [11], [17], [22]. The limited amputee subject pool presents a major challenge for developing new prosthesis control systems because it potentially places the disproportionately high burden of testing multiple critical control characteristics on a small subject population. While testing more able-bodied subjects improves statistical power, the relevance of such testing for amputees is limited given the substantial differences in neuromusculoskeletal condition between the two populations. Though our results are promising, more testing is needed to determine the viability of our model for clinical prosthesis control.

V. Conclusion

The observed trend of better real-time task performance with sEMG than iEMG was unexpected but highlighted important effects that electrode type may have on SPC of HMI. Differences in signal characteristics and user interface between sEMG and iEMG may explain the observed trend and should be investigated further. Our results contribute important clarity on the advantages and disadvantages of EMG electrode type that can inform development and implementation of SPC for clinical or research HMI applications.

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References

[1] C. Cipriani, F. Zaccone, S. Micera, and M. C. Carrozza, "On the shared control of an EMG-controlled prosthetic hand: Analysis of user–prosthesis interaction," *IEEE Trans. Robot.*, vol. 24, no. 1, pp. 170–184, Feb. 2008.

[2] M. Ison and P. Artemiadis, "The role of muscle synergies in myoelectric control: Trends and challenges for simultaneous multifunction control," *J. Neural Eng.*, vol. 11, no. 5, p. 051001, 2014.

[3] R. N. Scott, "Myoelectric control of prostheses and orthoses," *Bull. Prosthetics Res.*, pp. 93–114, 1967.

[4] B. Hudgins, P. Parker, and R. N. Scott, "A new strategy for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 40, no. 1, pp. 82–94, Jan. 1993.

[5] K. Englehart and B. Hudgins, "A robust, real-time control scheme for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 7, pp. 848–854, Jul. 2003.

[6] K. Davidge, "Multifunction myoelectric control using a linear electrode array," M.S. thesis, Dept. Elect. Comput. Eng., Univ. New Brunswick, Fredericton, NB, Canada, 1999.

[7] R. B. Stein *et al.*, "New approaches for the control of powered prostheses particularly by high-level amputees," *Bull. Prosthetics Res.*, vol. 17, no. 1, pp. 51–62, 1980.

[8] A. J. Young, L. H. Smith, E. J. Rouse, and L. J. Hargrove, "Classification of simultaneous movements using surface EMG pattern recognition," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 5, pp. 1250–1258, May 2013.

[9] J. J. Baker, E. Scheme, K. Englehart, D. T. Hutchinson, and B. Greger, "Continuous detection and decoding of dexterous finger flexions with implantable myoelectric sensors," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 18, no. 4, pp. 424–432, Aug. 2010.

[10] N. Jiang, K. B. Englehart, and P. A. Parker, "Extracting simultaneous and proportional neural control information for multiple-DOF prostheses from the surface electromyographic signal," *IEEE Trans. Biomed. Eng.*, vol. 56, no. 4, pp. 1070–1080, Apr. 2009.

[11] L. H. Smith, T. A. Kuiken, and L. J. Hargrove, "Evaluation of linear regression simultaneous myoelectric control using intramuscular EMG," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 4, pp. 737–746, Apr. 2016.

[12] D. Yatsenko, D. McDonnall, and K. S. Guillory, "Simultaneous, proportional, multi-axis prosthesis control using multichannel surface EMG," in *Proc. IEEE Annu. Int. Conf. Eng. Med. Biol. Soc.*, Aug. 2007, pp. 6134–6137.

[13] D. L. Crouch and H. Huang, "Lumped-parameter electromyogram-driven musculoskeletal hand model: A potential platform for real-time prosthesis control," *J. Biomech.*, vol. 49, no. 16, pp. 3901–3907, 2016.

[14] D. L. Crouch and H. H. Huang, "Musculoskeletal model-based control interface mimics physiologic hand dynamics during path tracing task," *J. Neural Eng.*, vol. 14, no. 3, p. 036008, Jun. 2017.

[15] D. Blana, E. K. Chadwick, A. J. van den Bogert, and W. M. Murray, "Real-time simulation of hand motion for prosthesis control," *Comput. Methods Biomech. Biomed. Eng.*, vol. 20, no. 5, pp. 540–549, 2017.

[16] L. Pan, D. Crouch, and H. Huang, "Musculoskeletal model for simultaneous and proportional control of 3-DOF hand and wrist movements from EMG signals," presented at the 8th Int. IEEE/EMBS Neural Eng. Conf., Shanghai, China, 2017.

[17] L. H. Smith and L. J. Hargrove, "Comparison of surface and intramuscular EMG pattern recognition for simultaneous wrist/hand motion classification," in *Proc. IEEE Annu. Int. Conf. Eng. Med. Biol. Soc.*, Jul. 2013, pp. 4223–4226.

[18] A. Fougner, E. Scheme, A. D. C. Chan, K. Englehart, and. Ø. Stavdahl, "Resolving the limb position effect in myoelectric pattern recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 19, no. 6, pp. 644–651, Dec. 2011.

[19] E. Scheme, A. Fougner, Ø. Stavdahl, A. D. C. Chan, and K. Englehart, "Examining the adverse effects of limb position on pattern recognition based myoelectric control," in *Proc. IEEE Annu. Int. Conf. Eng. Med. Biol.*, Aug./Sep. 2010, pp. 6337–6340.

[20] C. Cipriani, J. L. Segil, J. A. Birdwell, and R. F. Weir, "Dexterous control of a prosthetic hand using fine-wire intramuscular electrodes in targeted extrinsic muscles," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 4, pp. 828–836, Jul. 2014.

[21] J. A. Birdwell, L. J. Hargrove, R. F. Weir, and T. A. Kuiken, "Extrinsic finger and thumb muscles command a virtual hand to allow individual finger and grasp control," *IEEE Trans. Biomed. Eng.*, vol. 62, no. 1, pp. 218–226, Jan. 2015.

[22] L. H. Smith, T. A. Kuiken, and L. J. Hargrove, "Real-time simultaneous and proportional myoelectric control using intramuscular EMG," *J. Neural Eng.*, vol. 11, no. 6, p. 066013, Dec. 2014.

[23] R. F. Weir, P. R. Troyk, G. A. DeMichele, D. A. Kerns, J. F. Schorsch, and H. Maas, "Implantable myoelectric sensors (IMESs) for intramuscular electromyogram recording," *IEEE Trans. Biomed. Eng.*, vol. 56, no. 1, pp. 159–171, Jan. 2009.

[24] L. J. Hargrove, K. Englehart, and B. Hudgins, "A comparison of surface and intramuscular myoelectric signal classification," *IEEE Trans. Biomed. Eng.*, vol. 54, no. 5, pp. 847–853, May 2007.

[25] L. Pan, D. L. Crouch, and H. Huang, "Reliable musculoskeletalmodel-based EMG interface against upper limb posture changes," *J. Neuroeng. Rehabil.*, to be published.

[26] A. Phinyomark, S. Thongpanja, H. Hu, P. Phukpattaranont, and C. Limsakul, "The usefulness of mean and median frequencies in electromyography analysis," in *Computational Intelligence in Electromyography Analysis—A Perspective on Current Applications and Future Challenges*, G. R. Naik, Ed. London, U.K.: InTech, 2012.

[27] H. Christensen, K. Søgaard, B. R. Jensen, L. Finsen, and G. Sjøgaard, "Intramuscular and surface EMG power spectrum from dynamic and static contractions," *J. Electromyograph. Kinesiol.*, vol. 5, no. 1, pp. 27–36, Mar. 1995.

[28] A. Searle and L. Kirkup, "A direct comparison of wet, dry and insulating bioelectric recording electrodes," *Physiol. Meas.*, vol. 21, no. 2, pp. 271–283, May 2000.

[29] L. Hargrove, K. Englehart, and B. Hudgins, "The effect of electrode displacements on pattern recognition based myoelectric control," in *Proc. IEEE Int. Conf. Eng. Med. Biol. Soc.*, vol. 1, Aug./Sep. 2006, pp. 2203–2206.

[30] N. Jiang, H. Rehbaum, I. Vujaklija, B. Graimann, and D. Farina, "Intuitive, online, simultaneous, and proportional myoelectric control over two degrees-of-freedom in upper limb amputees," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 3, pp. 501–510, May 2014.

[31] G. Li, A. E. Schultz, and T. A. Kuiken, "Quantifying pattern recognition—Based myoelectric control of multifunctional transradial prostheses," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 18, no. 2, pp. 185–192, Apr. 2010.

[32] J. P. A. Dewald and R. F. Beer, "Abnormal joint torque patterns in the paretic upper limb of subjects with hemiparesis," *Muscle Nerve*, vol. 24, no. 2, pp. 273–283, Feb. 2001.

[33] C. Mercier, K. T. Reilly, C. D. Vargas, A. Aballea, and A. Sirigu, "Mapping phantom movement representations in the motor cortex of amputees," *Brain*, vol. 129, no. 8, pp. 2202–2210, Aug. 2006.

[34] M. A. Powell, R. R. Kaliki, and N. V. Thakor, "User training for pattern recognition-based myoelectric prostheses: Improving phantom limb movement consistency and distinguishability," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 3, pp. 522–532, May 2014.

[35] A. M. Simon, B. A. Lock, and K. A. Stubblefield, "Patient training for functional use of pattern recognition–controlled prostheses," *J. Prosthetics Orthotics*, vol. 24, no. 2, pp. 56–64, Apr. 2012.

[36] L. Resnik, K. Etter, S. L. Klinger, and C. Kambe, "Using virtual reality environment to facilitate training with advanced upper-limb prosthesis," *J. Rehabil. Res. Develop.*, vol. 48, no. 6, pp. 707–718, 2011.

[37] M. A. Powell and N. V. Thakor, "A training strategy for learning pattern recognition control for myoelectric prostheses," *J. Prosthetics Orthotics*, vol. 25, no. 1, pp. 30–41, Jan. 2013.

[38] A. W. Dromerick, C. N. Schabowsky, R. J. Holley, B. Monroe, A. Markotic, and P. S. Lum, "Effect of training on upper-extremity prosthetic performance and motor learning: A single-case study," *Arch. Phys. Med. Rehabil.*, vol. 89, no. 6, pp. 1199–1204, Jun. 2008.

[39] B. D. Farnsworth, R. J. Triolo, and D. J. Young, "Wireless implantable EMG sensing microsystem," presented at the SENSORS, Lecce, Italy, 2008.

[40] P. A. Lichter, E. H. Lange, T. H. Riehle, S. M. Anderson, and D. S. Hedin, "Rechargeable wireless EMG sensor for prosthetic control," in *Proc. IEEE Annu. Int. Conf. Eng. Med. Biol.*, Aug./Sep. 2010, pp. 5074–5076.

[41] A. M. Simon, L. J. Hargrove, B. A. Lock, and T. A. Kuiken, "The target achievement control test: Evaluating real-time myoelectric pattern recognition control of a multifunctional upper-limb prosthesis," *J. Rehabil. Res. Develop.*, vol. 48, no. 6, pp. 619–627, 2011.

[42] T. A. Kuiken *et al.*, "Targeted reinnervation for enhanced prosthetic arm function in a woman with a proximal amputation: A case study," *Lancet*, vol. 369, no. 9559, pp. 371–380, Feb. 2007.

[43] T. A. Kuiken *et al.*, "Targeted muscle reinnervation for real-time myoelectric control of multifunction artificial arms," *J. Amer. Med. Assoc.*, vol. 301, pp. 619–628, Feb. 2009.

[44] G. S. Dhillon and K. W. Horch, "Direct neural sensory feedback and control of a prosthetic arm," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 13, no. 4, pp. 468–472, Dec. 2005.

[45] M. Schiefer, D. Tan, S. M. Sidek, and D. J. Tyler, "Sensory feedback by peripheral nerve stimulation improves task performance in individuals with upper limb loss using a myoelectric prosthesis," *J. Neural Eng.*, vol. 13, no. 1, p. 016001, Feb. 2016.